



Performance Challenges for a Product Architect

Our previous article in this series discussed the choices available for an architect in platform selection. This article provides a conceptual overview of the performance parameters an architect needs to consider while creating a software-intensive system or solution. We also show ways to measure the quality of product architecture by categorising requirements as functional and non-functional. Once we understand how to look at the performance of a product, we can easily understand the different subsystems such as file systems, IPC and memory management. We then go on to discuss pragmatic ways of improving performance while creating the product.

There are different ways of looking at the quality of a product. The quality attributes or non-functional requirements of a product are categorised into three parts, as shown in Figures 1 through 7.

- Performance or efficiency requirements
- X-ability requirements
- Other non-functional requirements (structural and safety parameters)

X-ability requirements concern the availability and reliability of the product. The performance or efficiency requirements can be further divided into three sections (also see Reference 1):

- Software performance
 - Hardware performance
 - Other engineering domain performance parameters
- Operating system performance